Definição (EDM)

Seja $D = (D_{ij})$ uma matriz $n \times n$ com $D_{ii} = 0, D_{ji} = D_{ij}, D_{ij} \geq 0$, dizemos que D é uma matriz de distâncias Euclidianas (em inglês, EDM) se existem pontos $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^K$, onde $K \in \mathbb{Z}_+$, tal que

$$D(i,j) = D_{ij} = ||\mathbf{x}_i - \mathbf{x}_j||, \ \ i,j \in \{1, \ldots, n\}. \tag{1}$$

Observações:

- O menor K tal que exista $\mathbf{x} = (\mathbf{x}_1, \ldots, \mathbf{x}_n) : \ V \to \mathbb{R}^K$ e ocorra (1) é denominado a dimensão de D, $\dim(D) = K$.
- Se não existe tal $\mathbf{x}$ dizemos que $\dim(D) = \infty$.
- **Teorema.** *$D : n \times n$ com $D_{ii} = 0, D_{ji} = D_{ij}$ e $D_{ij} \geq 0$ é uma EDM $\iff$ existe $\{\mathbf{x}_i\}_{i=1}^n \subseteq \mathbb{R}^K$ e $\mathrm{I} = \{i_1, \ldots, i_{K+1}\} \subseteq \mathrm{I}_n$ tal que (a) $\mathbf{x}_{i_1} = 0$; (b) $\mathbf{x}_{i_j}(j-1) \neq 0$, se $j \in \mathrm{I}_{2,K+1}$; (c) $\mathbf{x}_{i_j}(i) = 0$, se $j \in \mathrm{I}_{2,K}$ e $i \in \mathrm{I}_{j,K}$.*

Algorithm 1 $K = \mathtt{edmsph}(D, x)$

1: $I = \{1, 2\}$
2: $K = 1$
3: $(x_1, x_2) = (0, \sqrt{D_{12}})$
4: **for** $i \in \{3, \ldots, n\}$ **do**
5: $\quad \Gamma = \bigcap_{j \in I} \mathbb{S}^K(x_j, D_{ij})$
6: $\quad$ **if** $\Gamma = \varnothing$ **then**
7: $\quad\quad$ **return** ∞
8: $\quad$ **else if** $\Gamma = \{p_i\}$ **then**
9: $\quad\quad x_i = p_i$
10: $\quad$ **else if** $\Gamma = \{p_i^+, p_i^-\}$ **then**
11: $\quad\quad x_i = p_i^+$
12: $\quad\quad x \leftarrow \mathrm{expand}(x)$
13: $\quad\quad I \leftarrow I \cup \{i\}$
14: $\quad\quad K \leftarrow K + 1$
15: $\quad$ **else**
16: $\quad\quad$ **error:** dim aff(span(x_I)) $< K - 1$
17: $\quad$ **end if**
18: **end for**
19: **return** K

Figura 1: Algoritmo que verifica se uma matriz completa é uma EDM.

Observações:

- O passo 5 tem custo $O(\dim(D)^2)$, pois $K \in \{2, \ldots, \dim(D)\}$.
- O passo 5 é a interseção de K esferas no $\mathbb{R}^K$ de centros A.I.
- Com a AGC tal interseção tem custo $O(K^2)$ e a qualidade das soluções é a mesma que no caso clássico.

Exemplo:

$$D = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \end{bmatrix}$$

- $n = 4$; $\mathrm{I} = \{1, 2\}$; $K = 1$; $(\mathbf{x}_1, \mathbf{x}_2) = (0, \sqrt{D_{12}}) = (0, 1)$
- $\underline{i = 3}$:
  $\Gamma = S^1(\mathbf{x}_1, D_{31}) \cap S^1(\mathbf{x}_2, D_{32}) = \{\mathbf{p}_3^+ = (1, 1), \mathbf{p}_3^- = (1, -1)\}$;
  $\mathbf{x}_3 := \mathbf{p}_3^+ = (1, 1)$; $\mathbf{x} \in \mathbb{R}^1 \mapsto \mathrm{expand}(\mathbf{x}) \in \mathbb{R}^2$;
  $\mathrm{I} = \{1, 2\} \mapsto \mathrm{I} = \{1, 2, 3\}$; $K = 1 \mapsto K = 2$
- $\underline{i = 4}$:
  $\Gamma = S^2(\mathbf{x}_1, D_{41}) \cap S^2(\mathbf{x}_2, D_{42}) \cap S^2(\mathbf{x}_3, D_{43}) = \{\mathbf{p}_4 = (0, 1, 0)\}$;
  $\mathbf{x}_4 := \mathbf{p}_4 = (0, 1, 0)$; $\mathbf{x} \in \mathbb{R}^2 \mapsto \mathrm{expand}(\mathbf{x}) \in \mathbb{R}^3$
  $\mathrm{I} = \{1, 2, 3\} \mapsto \mathrm{I} = \{1, 2, 3, 4\}$; $K = 2 \mapsto K = 3$
- *Solução:* $\{\mathbf{x}_1 = (0, 0), \mathbf{x}_2 = (1, 0), \mathbf{x}_3 = (1, 1), \mathbf{x}_4 = (0, 1)\} \subseteq \mathbb{R}^2$;
  $\dim(D) = 2$

Testes computacionais[1] **($D : n \times n$):**

n	LDE
5	0
6	0
7	2.08E-17
8	6.09E-17
9	3.8E-17
10	1.5E-17
20	2.9E-17
30	1.6E-17
40	0
50	1.37E-15
60	2.4E-18
70	0
80	0
90	-

Figura 2: Tabela com erros LDE.

Testes computacionais ($D : 6 \times 6$):

```
In[28]:= (*Define os valores dos paramateros e cria instancia com n atomos*)
n = 6;(*numero de atomos*)
dc = 10;(*valor de corte*)
{X, A, Dij, Dijs, Q, ind} = GeraInstancia[n, dc];
Print["D = ", MatrixForm[Dij]];
```

$$D = \begin{pmatrix} 0 & 1.526 & 2.49139 & 2.59018 & 2.24404 & 2.72602 \\ 1.526 & 0. & 1.526 & 2.49139 & 2.66973 & 2.60171 \\ 2.49139 & 1.526 & 0. & 1.526 & 2.49139 & 2.71703 \\ 2.59018 & 2.49139 & 1.526 & 0. & 1.526 & 2.49139 \\ 2.24404 & 2.66973 & 2.49139 & 1.526 & 0. & 1.526 \\ 2.72602 & 2.60171 & 2.71703 & 2.49139 & 1.526 & 0. \end{pmatrix}$$

```
In[32]:= {ksol2, solx2} = edmsph[Dij * Dij];
I = {1, 2}
K = 1
x = {{0, 0}, {1.526, 0}}
_____  i = 3  _____
> > I = {1, 2},  nI = Lenght[I] = 2
> > length[x] = 2, K = 1
```

$$> x = \begin{pmatrix} 0 & 1.526 \\ 0 & 0 \end{pmatrix}$$

```
> > Computa Γ = ⋂s^1(x_j, D_3j), j ∈ {1, 2}:
  >  n = 2,  m = kk = 1
```

$$> A = \begin{pmatrix} 0 & 1.526 \\ 0 & 0 \end{pmatrix} \quad d = \begin{pmatrix} 2.49139 \\ 1.526 \end{pmatrix}$$

```
  >  A = {{0, 1.526}, {0, 0}},  d = {{2.49139}, {1.526}}
  > δ = 2.07086
  > Γ = {{2.03376, 1.43905}, {2.03376, -1.43905}}
> > x_3 = {2.03376, 1.43905}
> > expand[x] = {{0, 0, 0}, {1.526, 0, 0}, {2.03376, 1.43905, 0}}
> > x_3 = {{0, 0, 0}, {1.526, 0, 0}, {2.03376, 1.43905, 0}}
> > I <-- I+1  =>  I = {1, 2, 3}
> > K <-- K+1  =>  K = 2
_____  i = 4  _____
> > I = {1, 2, 3},  nI = Lenght[I] = 3
> > length[x] = 3, K = 2
```

$$> x = \begin{pmatrix} 0 & 1.526 & 2.03376 \\ 0 & 0 & 1.43905 \\ 0 & 0 & 0 \end{pmatrix}$$

```
> > Computa Γ = ⋂s^2(x_j, D_4j), j ∈ {1, 2, 3}:
  >  n = 3,  m = kk = 2
```

$$> A = \begin{pmatrix} 0 & 1.526 & 2.03376 \\ 0 & 0 & 1.43905 \\ 0 & 0 & 0 \end{pmatrix} \quad d = \begin{pmatrix} 2.59018 \\ 2.49139 \\ 1.526 \end{pmatrix}$$

```
  >  A = {{0, 1.526, 2.03376}, {0, 0, 1.43905}, {0, 0, 0}},
     d = {{2.59018}, {2.49139}, {1.526}}
  > δ = 0.242245
  > Γ = {{0.927493, 2.36782, 0.492184}, {0.927493, 2.36782, -0.492184}}
> > x_4 = {0.927493, 2.36782, 0.492184}
> > expand[x] = {{0, 0, 0, 0}, {1.526, 0, 0, 0}, {2.03376, 1.43905, 0, 0},
                 {0.927493, 2.36782, 0.492184, 0}}
> > x_4 = {{0, 0, 0, 0}, {1.526, 0, 0, 0}, {2.03376, 1.43905, 0, 0},
          {0.927493, 2.36782, 0.492184, 0}}
> > I <-- I+1  =>  I = {1, 2, 3, 4}
> > K <-- K+1  =>  K = 3
```

Figura 3: Exemplo com o algoritmo `edmsph[]`.

Exemplo ($D : 6 \times 6$):

```
_____   i = 5   ________________________________________
> > I = {1, 2, 3, 4},   nI = Lenght[I] = 4
> > length[x] = 4, K = 3
 > x = {{0, 1.526, 2.03376, 0.927493}, {0, 0, 1.43905, 2.36782}, {0, 0, 0, 0.492184}, {0, 0, 0, 0}}
> > Computa Γ = ⋂s^3(x_j, D_5j), j ∈ {1, 2, 3, 4}:
  > n = 4,  m = kk = 3
  > A = {{0, 1.526, 2.03376, 0.927493}, {0, 0, 1.43905, 2.36782}, {0, 0, 0, 0.492184}, {0, 0, 0, 0}}   d = {{2.24404}, {2.66973}, {2.49139}, {1.526}}
  > δ = 0
  > Γ = {{0.0776387, 1.63995, 1.52979, 0}}
 > > x_5 = {{0, 0, 0, 0}, {1.526, 0, 0, 0}, {2.03376, 1.43905, 0, 0}, {0.927493, 2.36782, 0.492184, 0}, {0.0776387, 1.63995, 1.52979, 0}}
_____   i = 6   ________________________________________
> > I = {1, 2, 3, 4},   nI = Lenght[I] = 4
> > length[x] = 4, K = 3
 > x = {{0, 1.526, 2.03376, 0.927493, 0.0776387}, {0, 0, 1.43905, 2.36782, 1.63995}, {0, 0, 0, 0.492184, 1.52979}, {0, 0, 0, 0, 0}}
> > Computa Γ = ⋂s^3(x_j, D_6j), j ∈ {1, 2, 3, 4}:
  > n = 4,  m = kk = 3
  > A = {{0, 1.526, 2.03376, 0.927493}, {0, 0, 1.43905, 2.36782}, {0, 0, 0, 0.492184}, {0, 0, 0, 0}}   d = {{2.72602}, {2.60171}, {2.71703}, {2.49139}}
  > A = {{0, 1.526, 2.03376, 0.927493}, {0, 0, 1.43905, 2.36782}, {0, 0, 0, 0.492184}, {0, 0, 0, 0}},  d = {{2.72602}, {2.60171}, {2.71703}, {2.49139}}
  > δ = 0
  > Γ = {{0.979998, 0.788638, 2.41844, 0}}
> > x_6 = {{0, 0, 0, 0}, {1.526, 0, 0, 0}, {2.03376, 1.43905, 0, 0}, {0.927493, 2.36782, 0.492184, 0}, {0.0776387, 1.63995, 1.52979, 0}, {0.979998, 0.788638, 2.41844, 0}}
  . . . excluindo coordenada nula j = 4
_____________________________________________________________

> > Solução:
> > K = 3
   x_sol =
{{0, 0, 0}, {1.526, 0, 0}, {2.03376, 1.43905, 0}, {0.927493, 2.36782, 0.492184}, {0.0776387, 1.63995, 1.52979}, {0.979998, 0.788638, 2.41844}}
   x = {{0, 0, 0, 0}, {1.526, 0, 0, 0}, {2.03376, 1.43905, 0, 0}, {0.927493, 2.36782, 0.492184, 0}, {0.0776387, 1.63995, 1.52979, 0}, {0.979998, 0.788638, 2.41844, 0}}
3]:= Print["X_instância = ", MatrixForm[X]];
     Print["erro LDE = ", Style[erroLDE[solx2, Dij], Red]];

X_instância =
{{0, 0, 0}, {-1.526, 0., 0.}, {-2.03376, 1.43905, 0.}, {-0.927493, 2.36782, -0.492184}, {-0.0776387, 1.63995, -1.52979}, {-0.979998, 0.788638, -2.41844}}

erro LDE = 0.
```

Figura 4: Exemplo com o algoritmo `edmsph[]`.